

ROS 2 URDF of 4-DOF Industrial Arm

Title

URDF Modeling of a 4-DOF Industrial Robotic Arm in ROS 2

1. Problem Statement

Design and implement a **ROS 2 URDF model** of a 4-DOF industrial robotic arm that:

- Defines all links and joints correctly
- Includes visual and collision elements
- Uses proper joint types and limits
- Loads successfully in RViz
- Displays a correct TF tree

2. Tasks / Questions to Be Solved

Question 1: Robot Structure Design

Question: Define the kinematic structure of a 4-DOF industrial arm and label all joints and links.

Question 2: URDF Link Definition

Question: Create URDF definitions for each link, including visual and collision geometry.

Question 3: Joint Definition

Question: Define revolute joints with correct parent–child relationships, axes, and limits.

Question 4: Assembly Validation

Question: Assemble all links and joints into a complete URDF robot model.

Question 5: RViz Visualization

Question: Load the URDF model in RViz and verify correct visualization of the robot.

Question 6: TF Tree Verification

Question: Visualize and verify the TF tree of the robot using ROS 2 tools.

3. Tools to Be Used

- **ROS 2 (Humble / Iron)**
- **URDF / Xacro**
- **RViz2**
- **robot_state_publisher**

4. Submission Guidelines (Google Drive)

Upload the following to **Google Drive** and submit the shared link:

- **Project Report (PDF)**
- **URDF / Xacro Files**
- **Launch Files**
- **Screenshots of RViz and TF tree**

Folder Name:

ROS2_URDF_4DOF_Arm_<StudentName>

5. Evaluation Criteria (Both Projects)

- **Correctness of implementation**
- **Functional demonstration**
- **Code and file organization**
- **Clarity of results and documentation**

End of Problem Statements